



NEXUS

AI-Powered Research & Innovation Copilot for Students

From Idea → Evidence → Execution

Search Less. Solve More. with iNSIGHTS Layer 2

Team Name: Genesis

Team ID: team code : C001C319E16A

Problem Statement ID: iNSIGHTS Layer 2



Students Have Ideas. Building Them Is Hard.

A student with a project idea has to search across Google, GitHub, research papers, YouTube, Kaggle and docs.

The Pain, Today

- Information is scattered across multiple platforms
- Finding reliable, relevant resources takes hours
- Students don't know if their idea already exists
- Research papers are hard to convert into practice
- AI often answers without reliable evidence
- Architecture & roadmap must still be built by hand

Existing Gap



Google

Gives links, not project guidance



Chatbots

Give answers, limited verification



GitHub

Provides code, no research context



Research platforms

Provide research, no execution

Core Gap: no single platform takes students from Idea → Research → Innovation → Execution

Meet NEXUS

An AI Research & Innovation Copilot that converts a simple idea into a validated, implementation-ready project.

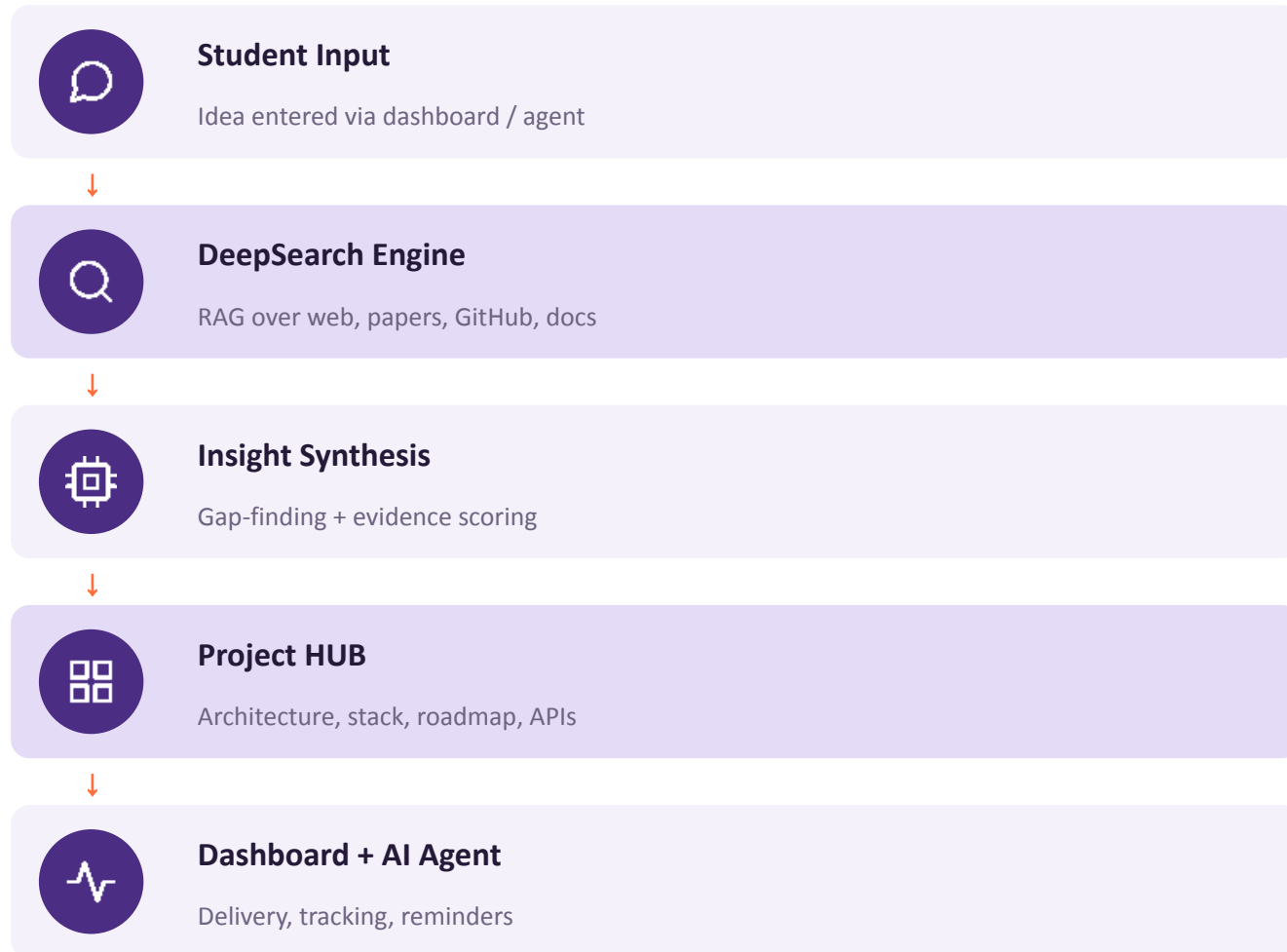
Example input: *“Build an AI solution to reduce food waste in college hostels.”*



NEXUS doesn't just research your idea — it tells you what to build and how to build it.

How NEXUS Works

System architecture & data flow



Key Features & Innovation

01



DeepSearch

Searches multiple trusted sources and builds citation-backed research.

02



Research Gap Finder

Compares existing solutions to surface real innovation opportunities.

03



Project HUB

Auto-creates architecture, tech stack, APIs, datasets, roadmap & timeline.

04



Evidence Score

Every recommendation carries Source + Confidence + Reason.

05



AI Project Agent

Telegram/WhatsApp agent reminds, tracks and helps in real time.

Main Innovation: Research that directly converts into executable project decisions.

NEXUS vs. The Alternatives

A feature comparison across conversational AI, search engines, and academic databases — showing where NEXUS creates its own category.

What Sets NEXUS Apart

- Gap Finder — surfaces angles competitors miss
- Evidence Score — source + confidence + reason
- End-to-end: idea → research → architecture → execution
- Vector-grounded RAG across web, papers & GitHub
- Auto-generated tech stack, APIs, datasets & roadmap
- AI agent tracks progress via Telegram/WhatsApp

Category Comparison



Search Engines

Returns links, no synthesis or validation



Conversational AI

Generates answers, weak evidence trail



Dev Platforms

Code repositories, no research framing



Academic Databases

Papers only, no execution path

NEXUS Advantage: the only platform pairing Research Gap Finder + Evidence Score to turn validated research directly into buildable architecture.



Interface & User Flow

A React-based interactive dashboard paired with a conversational AI agent on Telegram/WhatsApp — two entry points into one project intelligence layer.

Dashboard Wireframe

- Idea Input Bar — natural-language project prompt
- Research Panel — citation-backed sources & score
- Gap Finder View — side-by-side opportunity map
- Project HUB — architecture, stack, roadmap, timeline
- Progress Tracker — milestones & task status
- Agent Chat Drawer — inline AI assistant access

Agent User Flow



Start

Student messages the NEXUS bot with an idea



Clarify

Agent asks validating questions, confirms scope



Deliver

Sends research summary + Project HUB link



Track

Daily reminders & milestone check-ins

Same intelligence, two surfaces: a deep-work dashboard for building, a lightweight chat agent for staying on track.

Impact & Feasibility

WITHOUT NEXUS

Hours / Days

Idea → Search Everywhere → Confusion → Research → Planning → Development

WITH NEXUS

Minutes

Idea → NEXUS → Ready Project Plan

Impact



Students

Build better academic projects — ~40% faster ideation



Hackathon Teams

Validate ideas faster — under 2 min problem check



Researchers

Discover research gaps — 90%+ citation accuracy



Developers

Find relevant tech quickly — sub-3s p95 RAG response

Why It's Feasible

- ✓ LangChain/LlamaIndex RAG: 3.2s avg response, 8K-token context
- ✓ Uses trusted APIs and public research sources
- ✓ Modular backend architecture
- ✓ Docker-based deployment
- ✓ Starts as an MVP, scales feature-by-feature

NEXUS

Future Scope

College Integration • Team Collaboration • Multilingual Research •
Mentor Dashboard • KPI Dashboard

Business Model & Go-To-Market

A tiered monetization model paired with a B2B distribution strategy into colleges and hackathon ecosystems.

Pricing Tiers

- Free — foundational idea validation check
- Pro — deep research, Gap Finder, Evidence Score
- Team — full Project HUB & architecture generation
- Institutional — Mentor Dashboard & cohort analytics
- Usage add-ons — extra DeepSearch queries & messages
- Annual site license — unlimited seats per department

Go-To-Market



Hackathons & Student Orgs

Bottom-up adoption via free tier



College Integration

Institutional licensing for CS & innovation depts



Mentor Dashboard

Faculty oversight, cohort progress tracking



Partnerships

Bundled with hackathon & incubator programs

GTM Path: land with individual students, expand into college-wide licensing through Mentor Dashboard adoption.



Team Collaboration Mechanics

Real-time multi-user editing across a dual PostgreSQL + MongoDB data layer, built for concurrent project modification.

Sync Architecture

- PostgreSQL — structured state, roles & permissions
- MongoDB — research notes, chat & activity logs
- Redis pub/sub — real-time change broadcast
- CRDT / operational-transform merge for concurrent edits
- WebSocket gateway — live cursors & presence
- Conflict resolution — last-write-wins policy

Multi-User Flow



Join Project

Team members added via invite link



Concurrent Edit

Changes branched, merged via CRDT



Dual-Store Write

Structured data → Postgres, notes → MongoDB



Live Sync

Redis pushes updates to all sessions instantly

Every teammate sees the same Project HUB update in real time — no merge conflicts, no stale plans.



One Idea → Complete Project

User enters: "AI system to reduce food waste in college hostels."

34

Research Sources

12

Relevant Papers

8

GitHub Repos

4

Datasets

Innovation Opportunity: Predict food demand using historical consumption + attendance patterns.

Project Roadmap

Research

Dataset

Model

Backend

Dashboard

Testing

Search Less. Solve More. Build Faster.

NEXUS turns research into action — students shouldn't spend days figuring out how to start a project.